Pranav Dawange

Email: pranavdawange03@gmail.com | Phone: +91 7020917160
GitHub: github.com/Pranav-Dawange | LinkedIn: linkedin.com/in/pranav-dawange-935961329
Pune, Maharashtra, India



EDUCATION

Vishwakarma Institute of Technology, Pune **2024 – 2028**
B.Tech in Computer Science and Engineering (Data Science)

K.J. Somaiya College and K.B. Rohmare College **2022 – 2024**
Higher Secondary Certificate (HSC) – **87.0%**

Seva Niketan Convent School **2012 – 2022**
Secondary School Certificate (SSC) – **96.80%**

PROJECTS

Skin Lesion AI Analysis Platform | React 19, FastAPI, PyTorch, MongoDB, OpenCV, scikit-learn

- Developed a full-stack dermatological screening platform with JWT-authenticated FastAPI backend, MongoDB/GridFS storage, and a React frontend for image upload and clinical visualization.
- Fine-tuned a ResNet50 CNN on the HAM10000 dataset and implemented Grad-CAM from scratch for model explainability.
- Built a concurrent asynchronous pipeline (asyncio) combining classification, Grad-CAM, and GrabCut-based lesion segmentation.
- Engineered a metadata-driven risk-scoring layer using a scikit-learn Random Forest model.

Road Complaint Management System | FastAPI, PostgreSQL (Supabase), MongoDB, JWT, Pydantic, Bootstrap 5

- Built a role-based full-stack civic complaint platform with JWT authentication (citizen, officer, admin) and FastAPI REST APIs.
- Implemented Haversine-based geospatial auto-assignment to match complaints with nearest available field workers.
- Engineered PostgreSQL triggers, stored functions, and views for automated audit logging and ward-level analytics.
- Integrated background email notifications and on-demand PDF report generation using FastAPI BackgroundTasks.

AI-Powered Prescription Drug Interaction Detection System | Python, FastAPI, Node.js, Express.js, Next.js, MongoDB, YOLOv8, EasyOCR

- Designed and developed a microservice-based healthcare AI platform leveraging FastAPI, Express.js, and Next.js for automated prescription analysis.
- Implemented a computer vision pipeline using custom-trained YOLOv8 and EasyOCR to extract prescription information with enhanced accuracy.
- Engineered a risk-scoring algorithm using clinical indicators, fuzzy drug normalization, and interaction analysis to classify medication risk levels.
- Built secure REST APIs with JWT authentication and asynchronous processing for scalable end-to-end workflows.

ACHIEVEMENTS & PATENTS

- **Published Patent:** Official Patent published by relevant patent authority (**Application No. 202621043586**) | 2026
- **Finalist, InnoQuest 2025:** Selected for Final Round among participating engineering teams.
- **Second Round Qualifier, SSH 2026:** Advanced to Round 2 of the Symbiosis National Level Hackathon 2026.

OPEN SOURCE CONTRIBUTIONS

- **scikit-learn (Open Source ML Library):** Submitted pull request addressing a spurious EfficiencyWarning in DBSCAN clustering module for sorted precomputed graphs; followed contribution workflow with testing and linting.
- **ArogyaScript:** Contributed backend improvements through a merged GitHub pull request collaborating via version control workflows.

TECHNICAL SKILLS & CERTIFICATIONS

Languages: C, C++, Python, JavaScript, TypeScript
Frameworks & Libraries: React.js, Next.js, Node.js, Express.js, FastAPI, PyTorch, Scikit-learn, Pandas, NumPy, OpenCV
Databases & Cloud: MySQL, MongoDB, PostgreSQL (Supabase), Docker, Git, GitHub
Core CS Concepts: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, DBMS, Computer Networks
Certifications: Oracle Cloud Infrastructure 2025 Certified Data Science Professional • Google Data Analytics Professional Certificate • HackerRank Problem Solving (Intermediate) • AWS Cloud Practitioner (CLF-C02)